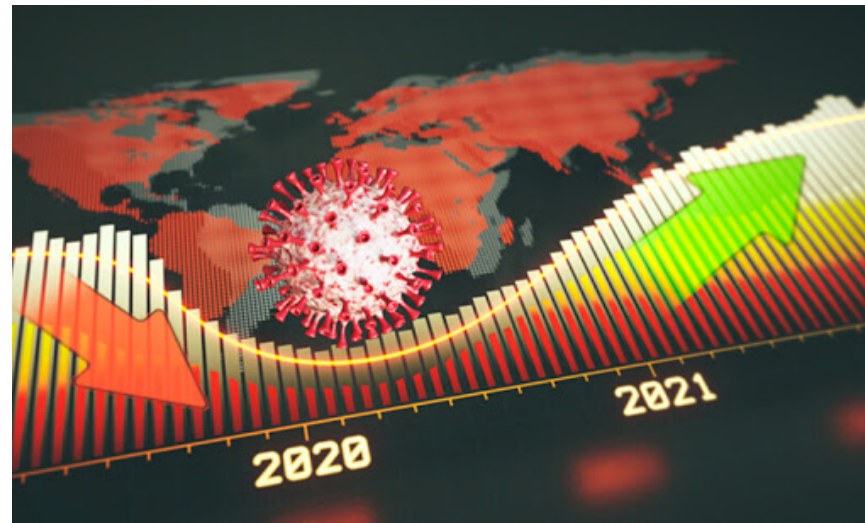


Will the anti-vaxxers kill us all?



Stacey Smith?

Department of Mathematics and Faculty of Medicine
The University of Ottawa



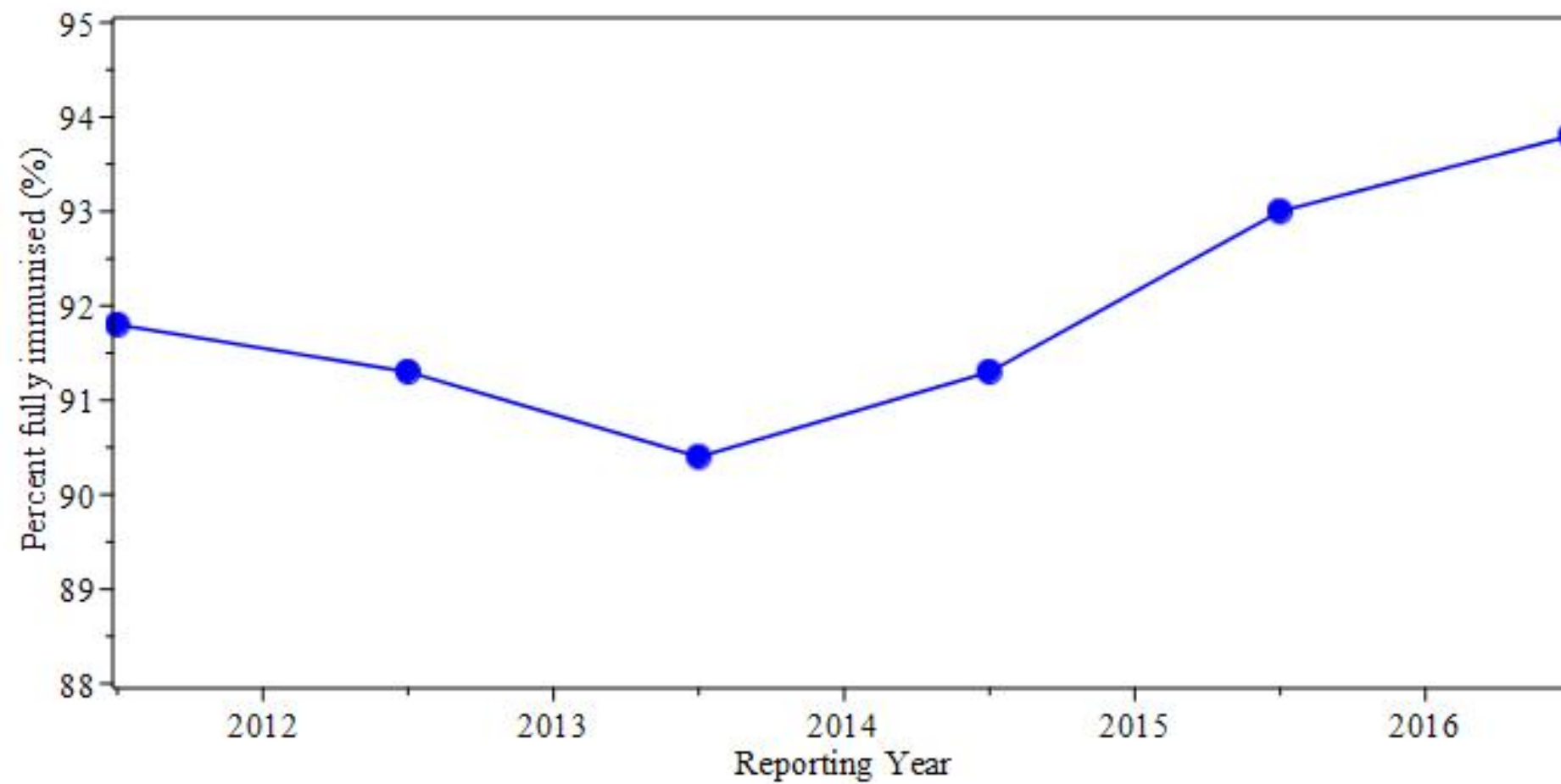


Figure 1.2: The percentage of fully immunised one year old children in Australia between 2011-12 and 2016-17. Generated from Australian Immunisation Register Data published by the Australian Institute of Health and Welfare [1].

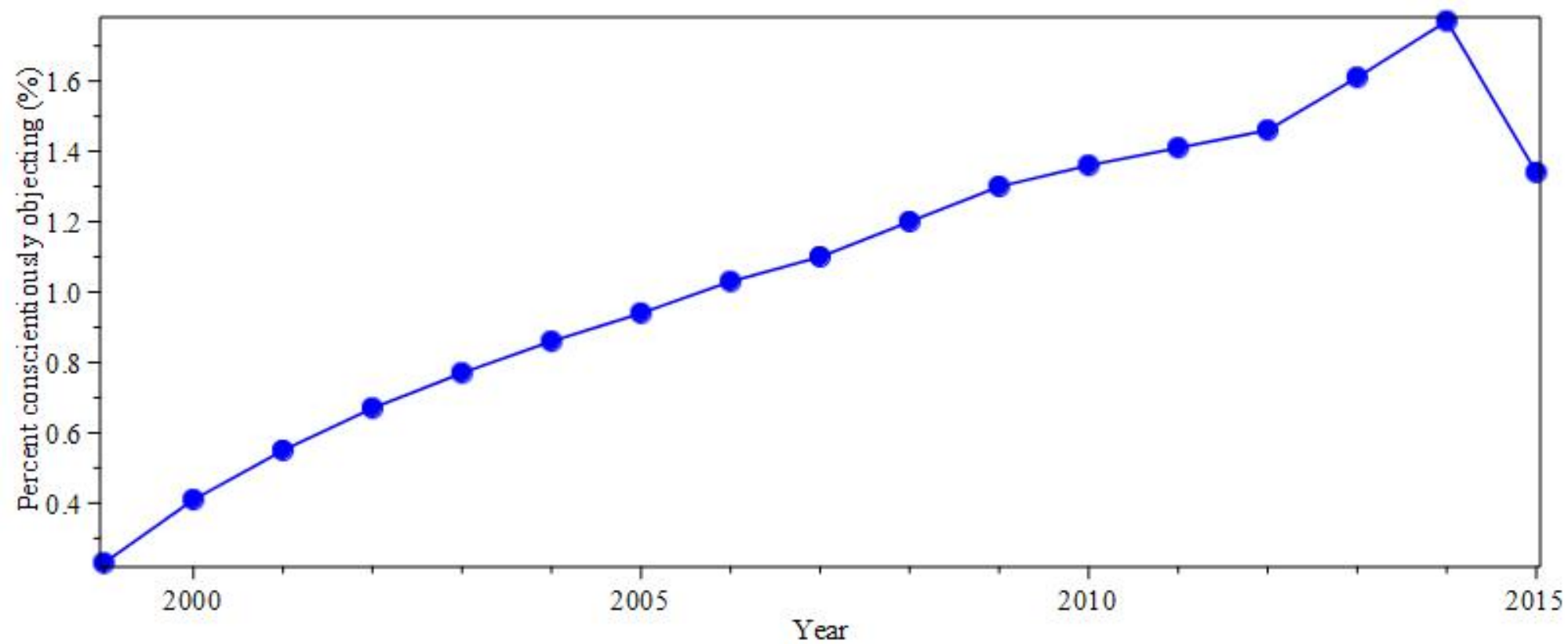
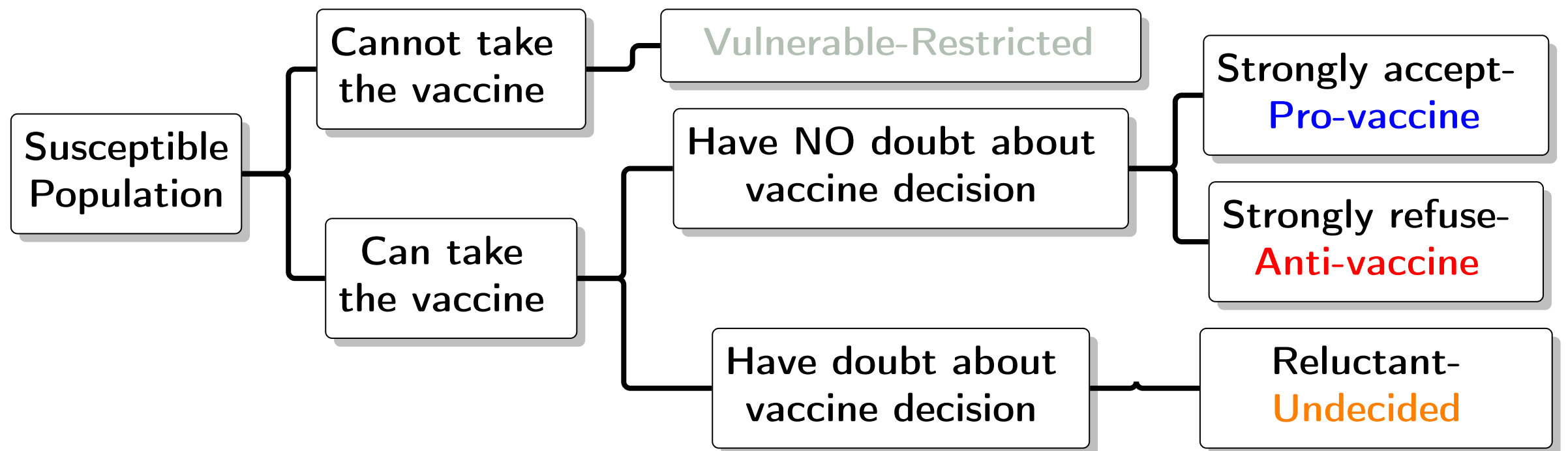


Figure 1.1: Australian Immunisation Register Data - year versus percentage of children with conscientious objection recorded. Data from Table 1.1.



Four groups

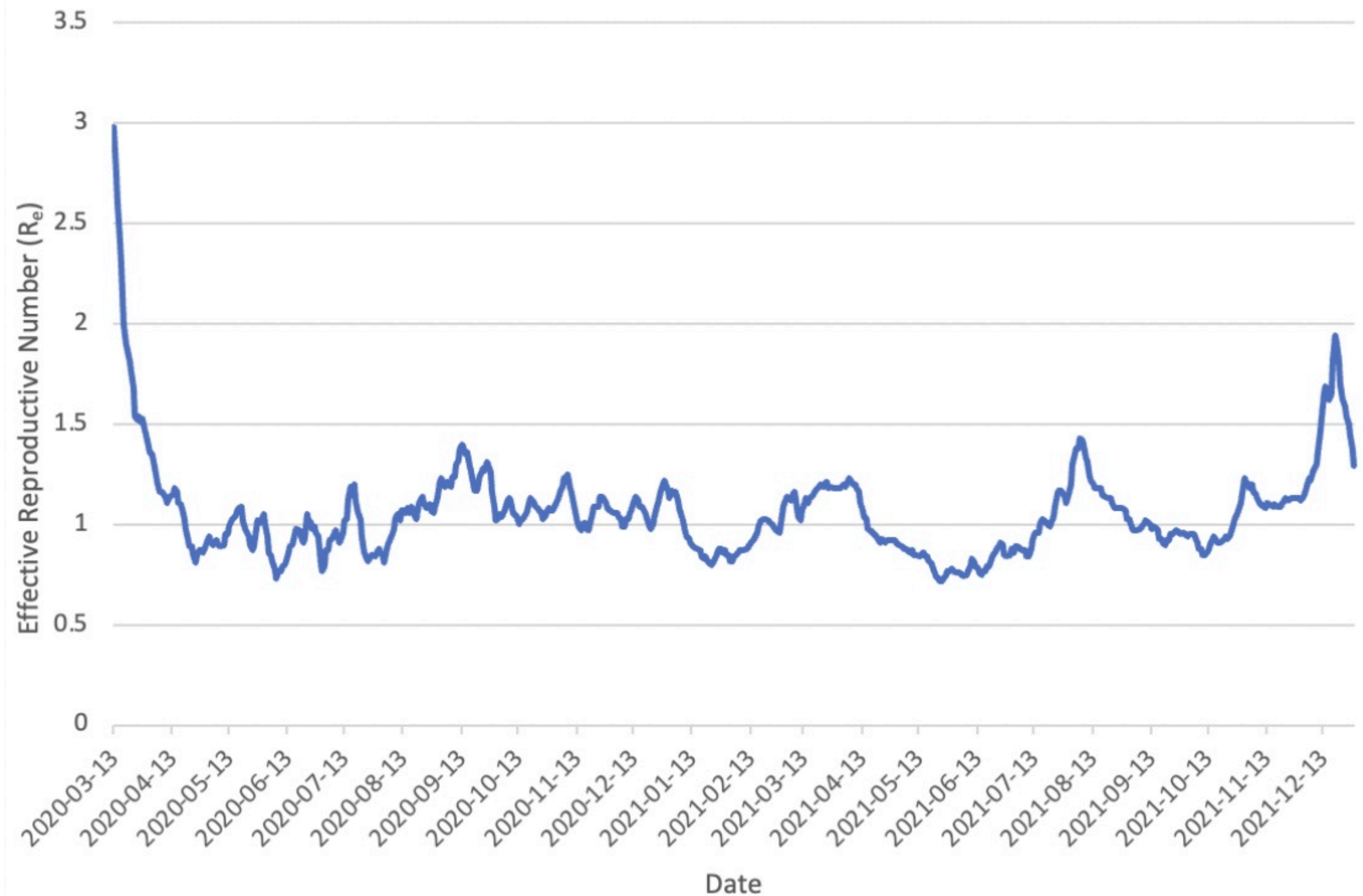
For any vaccine, there are four groups:

- a) those who never received the vaccine;
- b) those who received the vaccine but the vaccine did not take;
- c) those who received the vaccine, the vaccine took, but the vaccine waned over time; and
- d) those who received the vaccine, the vaccine took and for whom the vaccine did not wane over time.

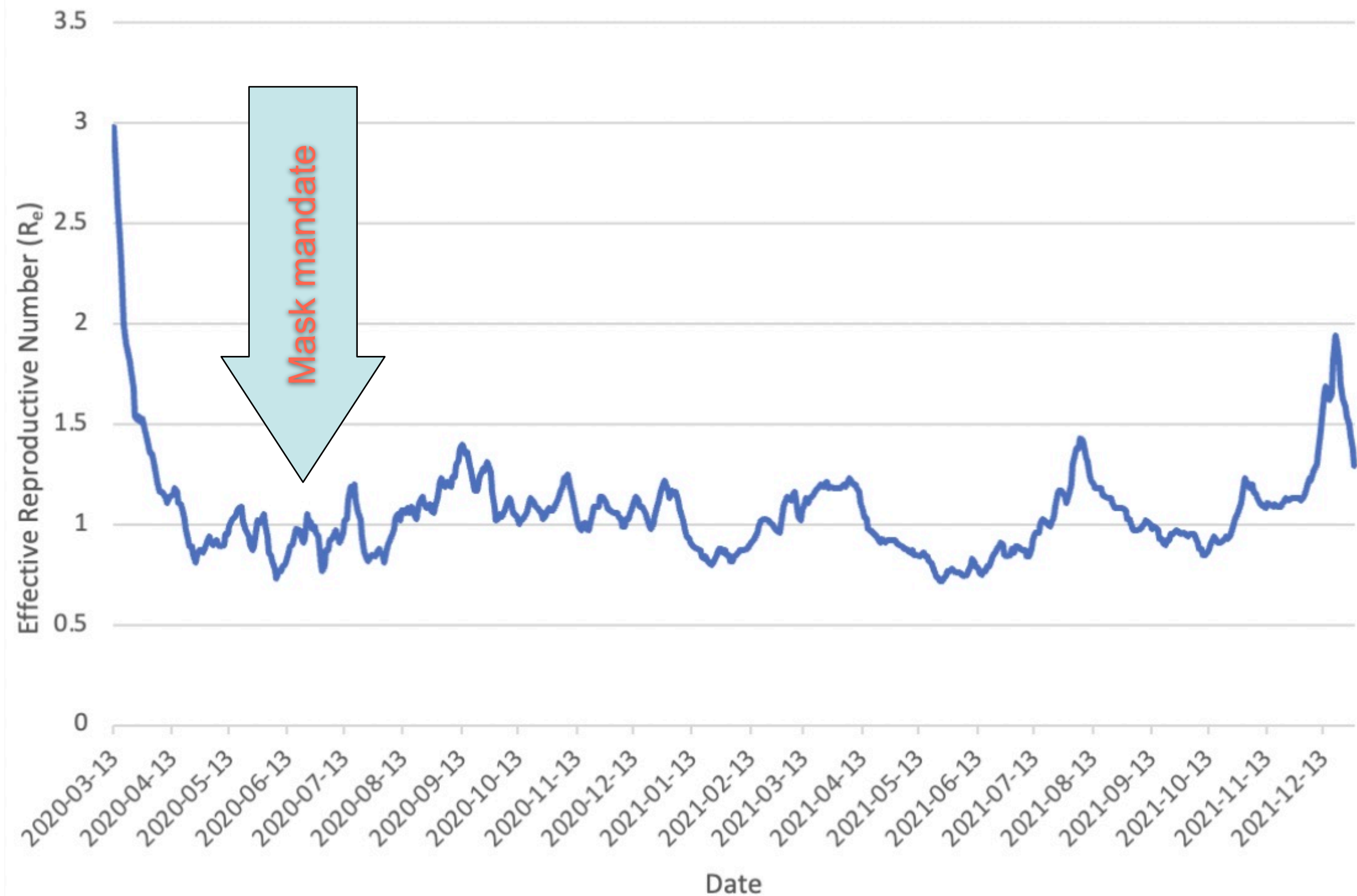
Risky behaviour

- ‘Unvaccinated’ individuals = groups (a)-(c)
- ‘Vaccinated’ individuals = group (d)
- Risky behaviour may increase or decrease for individuals in all groups.

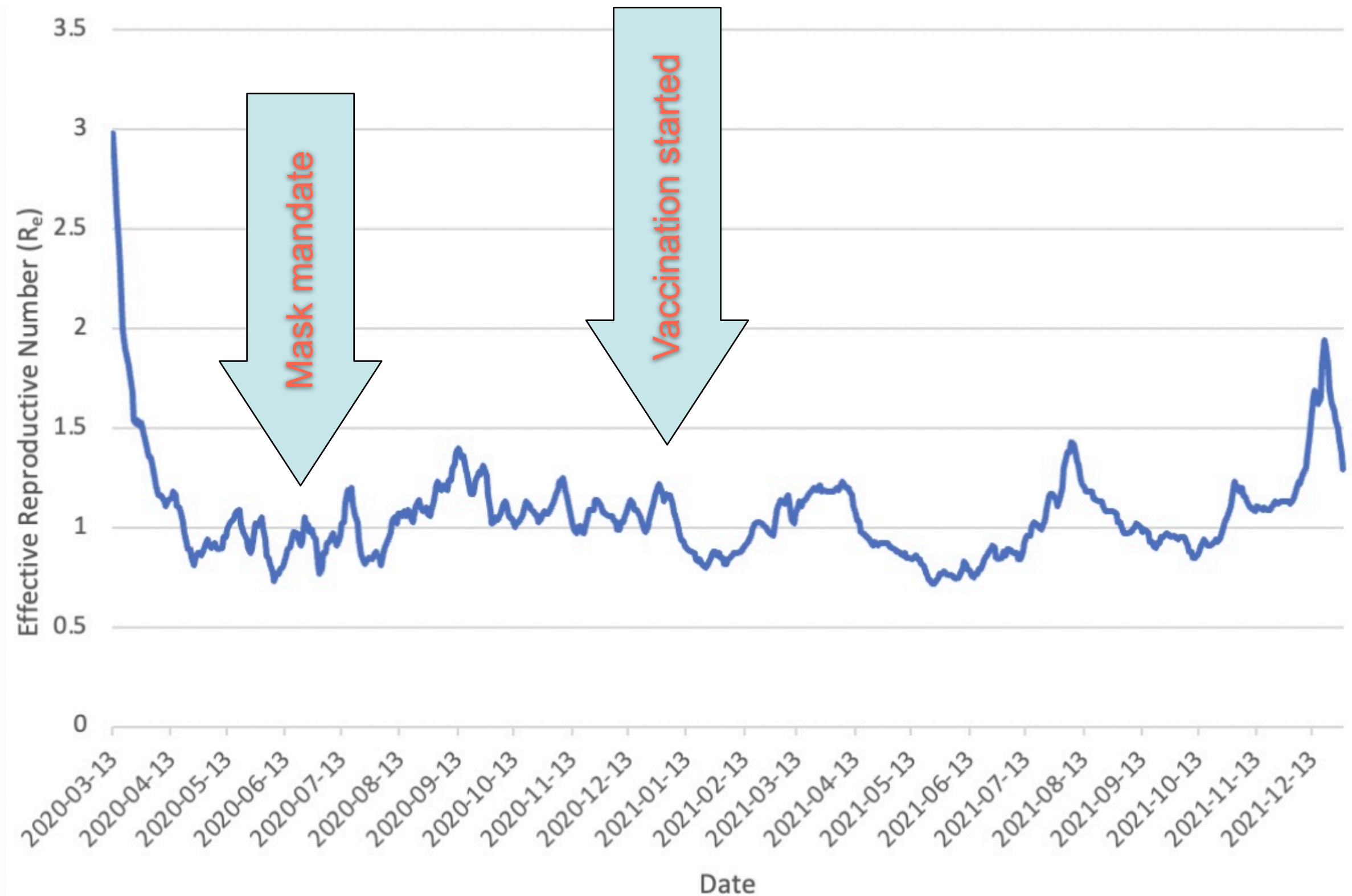
The reproduction number in Ontario



The reproduction number in Ontario

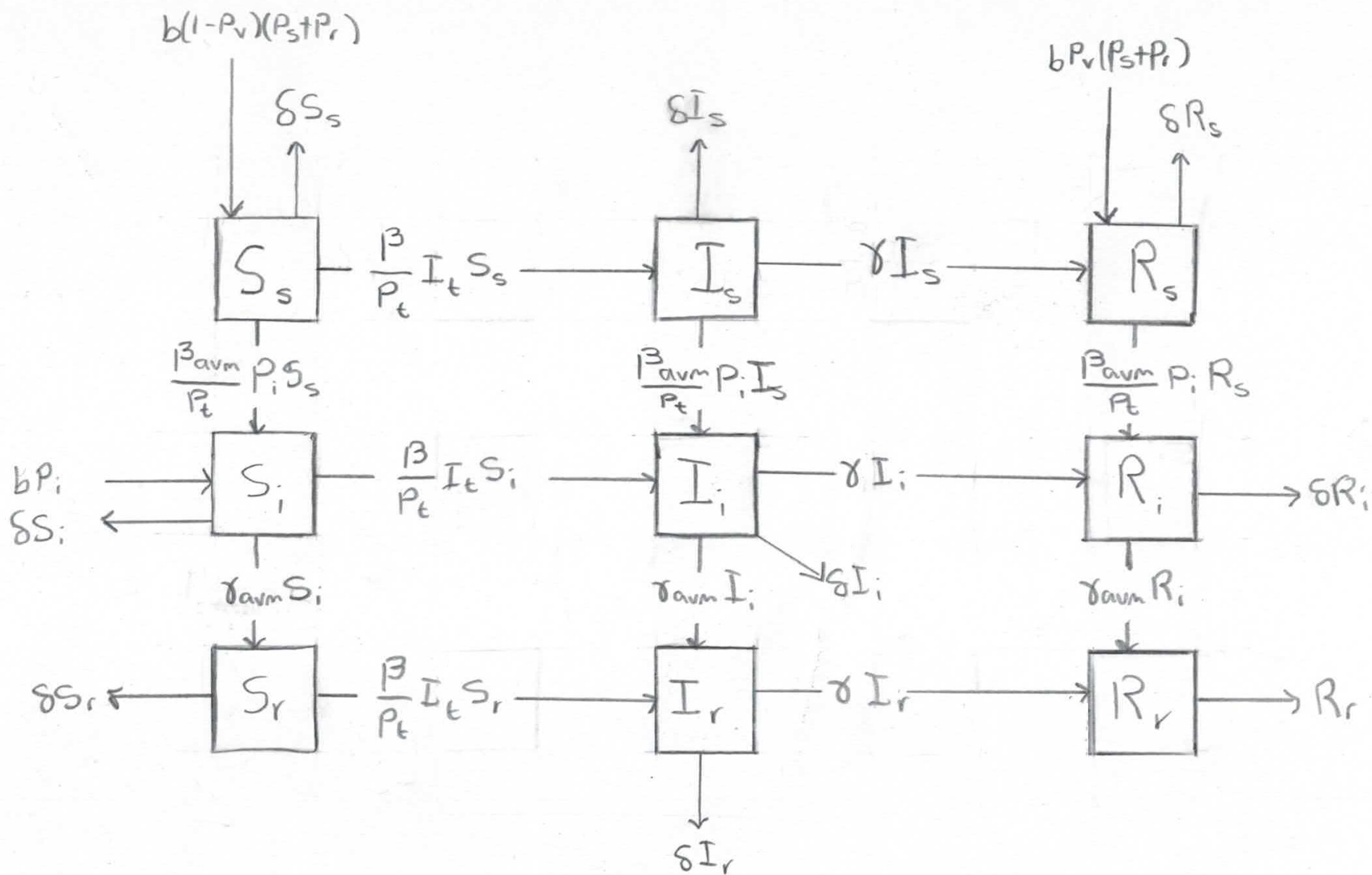


The reproduction number in Ontario



			Effectiveness against variants (E_{ij})				$\bar{E}_i$
	Vaccine type	Percentage (p_i)	Original strain	Alpha	Beta/Gamma	Delta	
Partial (single-dose) vaccination	Pfizer	5.40%	80%	66%	60%	56%	71%
	Moderna	2.20%	80%	83%	77%	72%	80%
	Vaxzevria	0.12%	70%	64%	48%	67%	65%
	Vaccine not reported	3.33%	92%	71%	65%	61%	74%
Full (two-doses) vaccination	Pfizer	28.12%	95%	89%	84%	87%	91%
	Moderna	8.01%	94%	92%	77%	72%	89%
	Vaxzevria	0.56%	85%	74%	48%	67%	75%
	Vaccine not reported	13.2%	95%	89%	82%	83%	90%
	mixed	9.7%	85%	74%	48%	67%	75%
Variant prevalence (π_j)		—	47%	31%	15%	6%	—

		Vaccines(Efficacy)						p_S^c	$\varepsilon(S)$	$\bar{E}_S$
	Strategy	$V_1(0.80)$	$V_2(0.85)$	$V_3(0.90)$	$V_4(0.94)$	$V_5(0.95)$	$V_6(0.95)$			
Single-vaccine strategies	S_{V_1}	1	0	0	0	0	0	0.750	0.000	0.80
	S_{V_2}	0	1	0	0	0	0	0.706	0.373	0.85
	S_{V_3}	0	0	1	0	0	0	0.667	0.704	0.90
	S_{V_4}	0	0	0	1	0	0	0.638	0.943	0.94
	S_{V_5}	0	0	0	0	1	0	0.632	1.000	0.95
	S_{V_6}	0	0	0	0	0	1	0.632	1.000	0.95
Multi-vaccine strategies	S_A	1/6	1/6	1/6	1/6	1/6	1/6	0.671	0.672	0.895
	S_B^1	0.13	0.15	0.17	0.18	0.19	0.19	0.667	0.691	0.898
	S_B^{10}	0.04	0.08	0.15	0.23	0.25	0.25	0.650	0.850	0.924
	S_B^{20}	0.01	0.03	0.10	0.25	0.30	0.30	0.640	0.926	0.937
	S_C^1	0.12	0.15	0.17	0.18	0.19	0.19	0.665	0.722	0.903
	S_C^{10}	0.01	0.03	0.11	0.25	0.30	0.30	0.640	0.926	0.937
	S_C^{20}	0.00	0.00	0.05	0.25	0.35	0.35	0.635	0.966	0.944
	S_D^1	0.05	0.10	0.14	0.19	0.26	0.26	0.650	0.838	0.922
	S_D^{10}	0.00	0.00	0.00	0.02	0.49	0.49	0.632	1.000	0.950
	S_D^{20}	0.00	0.00	0.00	0.00	0.50	0.50	0.632	1.000	0.950



Developing risk equations

- β' = probability of transmission with a given protection type



Developing risk equations

- β' = probability of transmission with a given protection type
- $(1-\beta')$ = probability of remaining uninfected



Developing risk equations

- β' = probability of transmission with a given protection type
- $(1-\beta')$ = probability of remaining uninfected
- Thus the probability of COVID acquisition is



Developing risk equations

- β' = probability of transmission with a given protection type
- $(1-\beta')$ = probability of remaining uninfected
- Thus the probability of COVID acquisition is

$$\text{Risk} = 1 - \prod_{\substack{\text{all protection} \\ \text{protocols}}}^i [1 - \beta']^{p_i}$$



Developing risk equations

- β' = probability of transmission with a given protection type
- $(1-\beta')$ = probability of remaining uninfected
- Thus the probability of COVID acquisition is

$$\text{Risk} = 1 - \prod_{\substack{\text{all protection} \\ \text{protocols}}}^i [1 - \beta']^{p_i}$$

where p_i is the proportion of times a given protection protocol is employed



Developing risk equations

- β' = probability of transmission with a given protection type
- $(1-\beta')$ = probability of remaining uninfected
- Thus the probability of COVID acquisition is

$$\text{Risk} = 1 - \prod_{\substack{\text{all protection} \\ \text{protocols}}}^i [1 - \beta']^{p_i}$$

where p_i is the proportion of times a given protection protocol is employed – including no protection.



Pre-vaccination risk

- Protection options: handwashing, masks, distancing or nothing



Pre-vaccination risk

- Protection options: handwashing, masks, distancing or nothing
- The daily risk is thus



Pre-vaccination risk

- Protection options: handwashing, masks, distancing or nothing
- The daily risk is thus

$$r_1 = 1 - [1 - (1 - e_D)\beta]^{p_0} [1 - (1 - e_M)\beta]^{p_1} [1 - (1 - e_H)\beta]^{p_2} \times \\ [1 - (1 - e_D)(1 - e_M)\beta]^{p_3} [1 - (1 - e_D)(1 - e_H)\beta]^{p_4} \times \\ [1 - (1 - e_M)(1 - e_H)\beta]^{p_5} [1 - (1 - e_D)(1 - e_M)(1 - e_H)\beta]^{p_6} [1 - \beta]^{1 - \sum_{i=0}^6 p_i}$$

β =transmission
 p_j =proportion of times
protection is used



Pre-vaccination risk

- Protection options: handwashing, masks, distancing or nothing
- The daily risk is thus

$$r_1 = 1 - [1 - (1 - e_D)\beta]^{p_0} [1 - (1 - e_M)\beta]^{p_1} [1 - (1 - e_H)\beta]^{p_2} \times \\ [1 - (1 - e_D)(1 - e_M)\beta]^{p_3} [1 - (1 - e_D)(1 - e_H)\beta]^{p_4} \times \\ [1 - (1 - e_M)(1 - e_H)\beta]^{p_5} [1 - (1 - e_D)(1 - e_M)(1 - e_H)\beta]^{p_6} [1 - \beta]^{1 - \sum_{i=0}^6 p_i}$$

- e_j =efficacy of intervention method.

β =transmission
 p_j =proportion of times
protection is used



Post-vaccine risk

- The daily risk with a vaccine will be

Post-vaccine risk

- The daily risk with a vaccine will be

$$\begin{aligned} r_2 = & 1 - \left[1 - (1 - e_D)\beta\right]^{q_0} \left[1 - (1 - e_M)\beta\right]^{q_1} \left[1 - (1 - e_H)\beta\right]^{q_2} \left[1 - (1 - e_D)(1 - e_M)\beta\right]^{q_3} \times \\ & \left[1 - (1 - e_D)(1 - e_H)\beta\right]^{q_4} \left[1 - (1 - e_M)(1 - e_H)\beta\right]^{q_5} \left[1 - (1 - e_D)(1 - e_M)(1 - e_H)\beta\right]^{q_6} \times \\ & \left[1 - (1 - e_V)\beta\right]^{q_7} \left[1 - (1 - e_V)(1 - e_D)\beta\right]^{q_8} \left[1 - (1 - e_V)(1 - e_M)\beta\right]^{q_9} \times \\ & \left[1 - (1 - e_V)(1 - e_H)\beta\right]^{q_{10}} \left[1 - (1 - e_V)(1 - e_D)(1 - e_M)\beta\right]^{q_{11}} \times \\ & \left[1 - (1 - e_V)(1 - e_D)(1 - e_H)\beta\right]^{q_{12}} \left[1 - (1 - e_V)(1 - e_M)(1 - e_H)\beta\right]^{q_{13}} \left[1 - \beta\right]^{1 - \sum_{i=0}^{13} q_i} \end{aligned}$$

*β =transmission e_j =efficacy of protection
 q_j =proportion of times protection is used*

Post-vaccine risk

- The daily risk with a vaccine will be

$$r_2 = 1 - [1 - (1 - e_D)\beta]^{q_0} [1 - (1 - e_M)\beta]^{q_1} [1 - (1 - e_H)\beta]^{q_2} [1 - (1 - e_D)(1 - e_M)\beta]^{q_3} \times \\ [1 - (1 - e_D)(1 - e_H)\beta]^{q_4} [1 - (1 - e_M)(1 - e_H)\beta]^{q_5} [1 - (1 - e_D)(1 - e_M)(1 - e_H)\beta]^{q_6} \times \\ [1 - (1 - e_V)\beta]^{q_7} [1 - (1 - e_V)(1 - e_D)\beta]^{q_8} [1 - (1 - e_V)(1 - e_M)\beta]^{q_9} \times \\ [1 - (1 - e_V)(1 - e_H)\beta]^{q_{10}} [1 - (1 - e_V)(1 - e_D)(1 - e_M)\beta]^{q_{11}} \times \\ [1 - (1 - e_V)(1 - e_D)(1 - e_H)\beta]^{q_{12}} [1 - (1 - e_V)(1 - e_M)(1 - e_H)\beta]^{q_{13}} [1 - \beta]^{1 - \sum_{i=0}^{13} q_i}$$

- We did not include the combination of all four protection options in r_2

β =transmission e_j =efficacy of protection
 q_j =proportion of times protection is used

Post-vaccine risk

- The daily risk with a vaccine will be

$$r_2 = 1 - [1 - (1 - e_D)\beta]^{q_0} [1 - (1 - e_M)\beta]^{q_1} [1 - (1 - e_H)\beta]^{q_2} [1 - (1 - e_D)(1 - e_M)\beta]^{q_3} \times \\ [1 - (1 - e_D)(1 - e_H)\beta]^{q_4} [1 - (1 - e_M)(1 - e_H)\beta]^{q_5} [1 - (1 - e_D)(1 - e_M)(1 - e_H)\beta]^{q_6} \times \\ [1 - (1 - e_V)\beta]^{q_7} [1 - (1 - e_V)(1 - e_D)\beta]^{q_8} [1 - (1 - e_V)(1 - e_M)\beta]^{q_9} \times \\ [1 - (1 - e_V)(1 - e_H)\beta]^{q_{10}} [1 - (1 - e_V)(1 - e_D)(1 - e_M)\beta]^{q_{11}} \times \\ [1 - (1 - e_V)(1 - e_D)(1 - e_H)\beta]^{q_{12}} [1 - (1 - e_V)(1 - e_M)(1 - e_H)\beta]^{q_{13}} [1 - \beta]^{1 - \sum_{i=0}^{13} q_i}$$

- We did not include the combination of all four protection options in r_2
 - few people will go to such extremes once a vaccine is available.

*β =transmission e_j =efficacy of protection
 q_j =proportion of times protection is used*